

MOHAMMAD AFAQ

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mafaq

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Oulu, Finland

SUMMARY

Software Engineer with strong hands-on experience in C/C++, modern C++ (C++14/C++20), Python, embedded systems, and Yocto-based build environments. Skilled in developing reliable low-level software, SDK maintenance, target testing, debugging, and code optimization with a focus on performance, efficiency, and security. Experienced with CI/CD automation, Robot Framework, GoogleTest, Jenkins, Linux/WSL development environments, and cross-functional collaboration in software projects.

EXPERIENCE

Elektrobit Automotive Finland Oy

Software Engineer / Design Lead

Oct 2022 – Present

Oulu, Finland

- Lead the design and technical ownership of two **Adaptive AUTOSAR components** for Linux-based automotive platform.
- Developed production-ready **C++ Adaptive applications** with focus on security, reliability, maintainability, and performance.
- Improved software quality through code coverage **{BullsEye}**, and unit, integration, and fuzz testing using **{GoogleTest, GoogleMock, and Robot Framework}**.
- Leveraged AI-assisted tools to accelerate debugging, implementation, and development workflows.
- Coordinated with multiple component owners to maintain requirements, resolve integration issues, and support software development across the build system.

Teradata Global Consulting

Associate Consultant

Sept 2021 – Aug 2022

Islamabad, Pakistan

- Supported enterprise **ETL/DCM systems** for clients in the retail sector.
- Managed SQL database operations including performance tuning, monitoring, backup, and recovery.
- Supported cloud infrastructure and integration workflows using **Microsoft Azure**.
- Automated operational tasks and service workflows to improve efficiency and reduce manual effort.

Educative Inc

Technical Engineer

June 2021 – Sep 2021

Lahore, Pakistan

- Docker & Kubernetes
- Shell Scripting & Elixir Testing

EDUCATION

M.Sc. (Technology) in Computing Sciences

TAMPERE UNIVERSITY

Aug 2022 - Mar 2026

Tampere, Finland

Major: Signal Processing and Machine Learning

Relevant Coursework: Machine Learning, Artificial Intelligence, Pattern Recognition, Advanced Signal Processing, Computational Diagnostics of Data, Complex Networks, Functional Programming, Programming Techniques in C

B.Sc. in Electrical Engineering

LAHORE UNIVERSITY OF MANAGEMENT SCIENCES

Sep 2017 - May 2021

Lahore, Pakistan

Relevant Coursework: Devices and Electronics, Microcontrollers, Advanced Digital Signal Processing, Deep Learning, Speech Processing, Applied Probability, Econometrics

TECHNICAL STRENGTHS

Programming Languages	C, C++{C++14, C++20}, Python, SQL, Shell Scripting
Operating Systems & Environments	Linux, Ubuntu, WSL, VirtualBox, Embedded Linux
Development & Build Tools	Git, GitHub, Docker, Yocto, Jenkins, CI/CD Pipelines, Makefiles, GCC, CMake
Testing & Validation	Robot Framework, GoogleTest, GoogleMock, Target Testing, Debugging, Unit Testing, Integration Testing, Fuzz Testing, GDB
Embedded & Automotive Software	Embedded Systems, SDK Maintenance, AUTOSAR, Secure Software Development
Software Engineering Practices	Agile/Scrum, Jira, Version Control, Code Reviews, Requirement Tracing, Code Coverage via Bullsye SwDD and SwAD Technical Documentation
AI & Automation	AI-assisted Development, GitHub CoPilot, Agent led Development
Professional Skills	Customer Support, Cross-Component Support, Problem-Solving, Critical Thinking, Team Collaboration

PROJECTS

Elektrobit Automotive Finland Oy

Volkswagen ICASr Program

- Developed and maintained embedded C++ ({C++14, C++20}) software components for Volkswagen's ICASr software-defined vehicle platform
- Implemented new features and Improvements while maintaining existing functionality across multiple software components
- Supporting migration from C++14 to C++20 standards.
- Linux platform updates, SDK upgrades, and software stack maintenance to ensure compatibility and long-term support

Master's Thesis

Model Quantization for Efficient Differential Mobility Spectrometry (DMS) Analysis

- Researched and evaluated multiple attention mechanisms to improve DMS plot prediction using a U-Net-based architecture
- Investigated model quantization techniques and analyzed their impact on accuracy, memory footprint, and inference performance for embedded deployment

Bachelor's Thesis

Detection and Removal of Motion Artifacts from Photoplethysmogram {PPG} Data

- Processed and analyzed PPG sensor data affected by motion and perfusion artifacts.
- Implemented real-time overlapping window based algorithm for artifact removal and improved heart rate accuracy.

CERTIFICATIONS

- CMake for Cross-Platform C++ Project Building
- Teradata Vantage Associate Certified
- Microsoft Azure Fundamentals Certified
- 2023 Web Development Bootcamp
- C++ Unit Testing: Google Test and Google Mock

REFERENCES

- Available Upon Request